

CS & IT ENGINEERING



Theory of Computation

DFA

Lecture No.- 09



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Topics to be covered



Topic

$NFA \rightarrow DFA$

Topic

$\textcircled{n}$ $\xrightarrow{\text{ } n \text{ states}}$
 $E-NFA \rightarrow NFA$

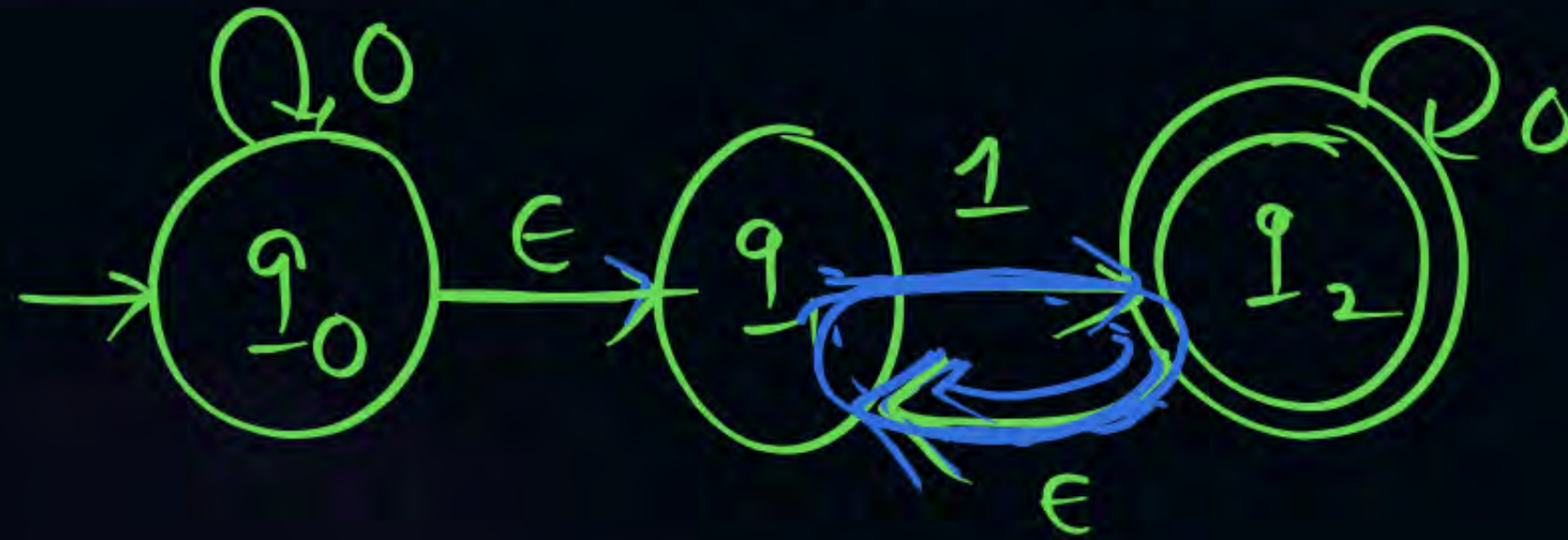
$E-NFA \rightarrow \textcircled{\text{min DFA}}$

n $\textcircled{\text{min } 1 \dots \text{max } 2^n}$



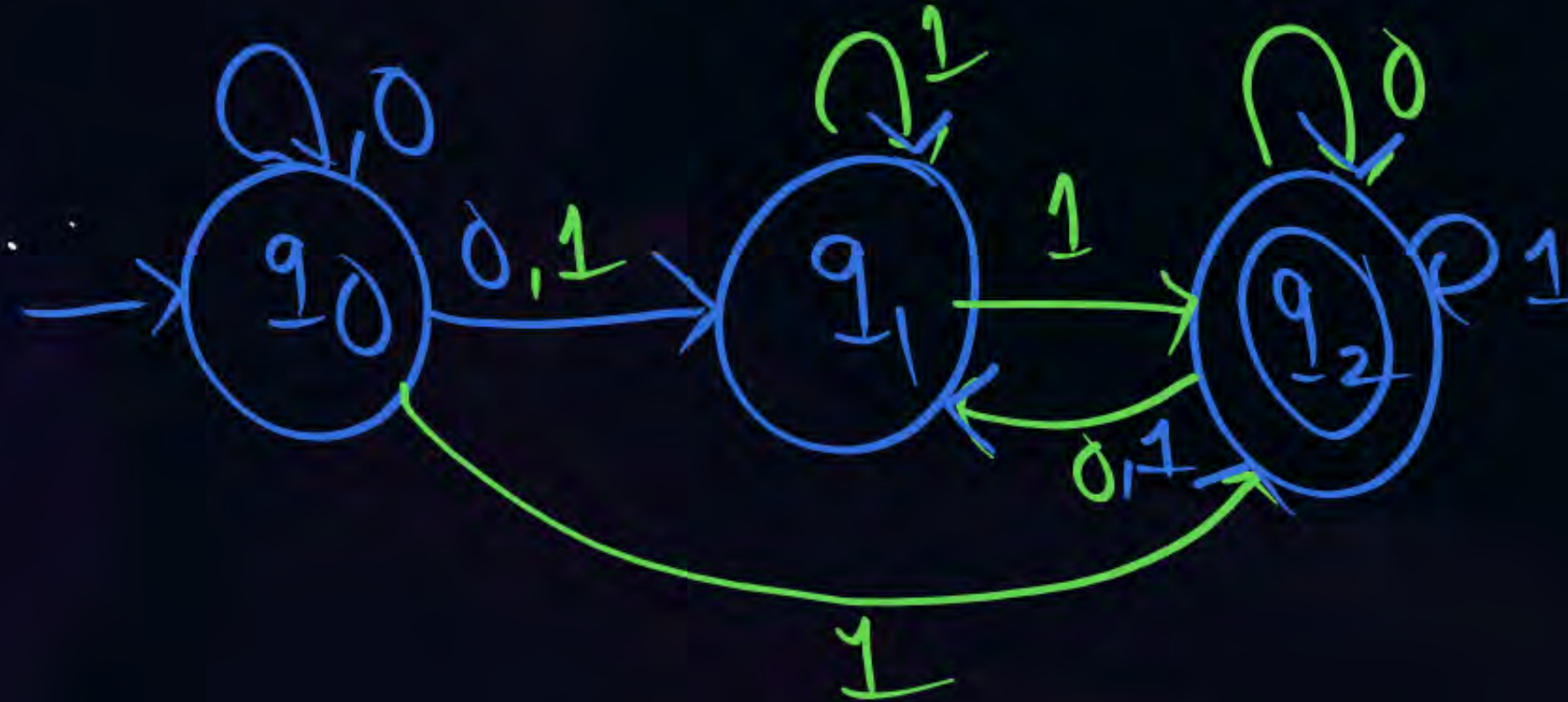
(Q) Construct NFA for given E-NFA

E-NFA



$\epsilon \cdot 1 \cdot \epsilon = 1$

NFA



$$\delta(q_0, 0) = \{q_0, q_1\}$$

$$\delta(q_0, 1) = \{q_2, q_1\}$$

$$\delta(q_1, 0) = \emptyset$$

$$\delta(q_1, 1) = \{q_2, q_1\}$$

$$\delta(q_2, 0) = \{q_1, q_2\}$$

$$\delta(q_2, 1) = \{q_1, q_2\}$$

(Q) Construct min DFA for given E-NFA.

E-NFA



NFA

	a	b	c
q0	{q0, q1, q2}	{q1, q2}	q2
q1	—	{q1, q2}	q2
q2	—	—	q2

DFA

	a	b	c
q0	[q0, q1, q2]	[q1, q2]	q2
q1	[q0, q1, q2]	[q1, q2]	q2
q2	[q1, q2]	[q1, q2]	q2
q3	q3	q3	q3

4 States

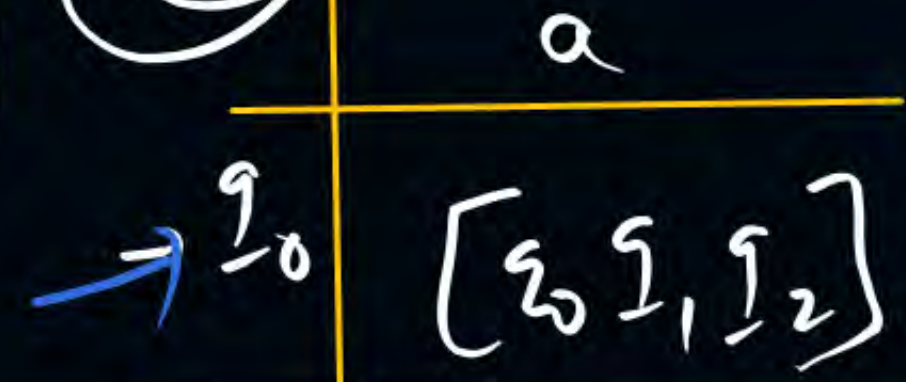
Q) How many states in min DFA for given E-NFA

E-NFA

NFA

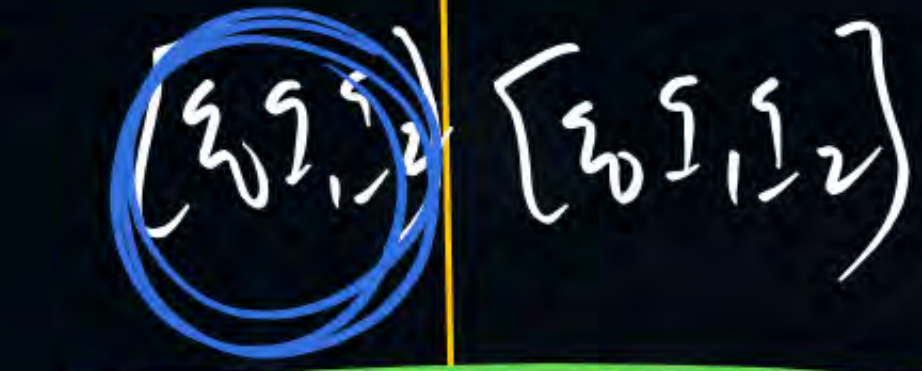
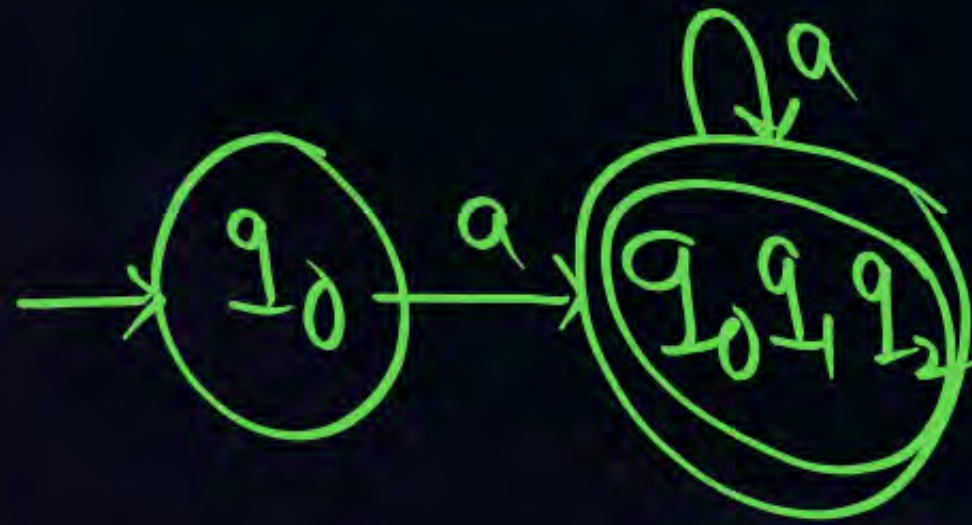


DFA



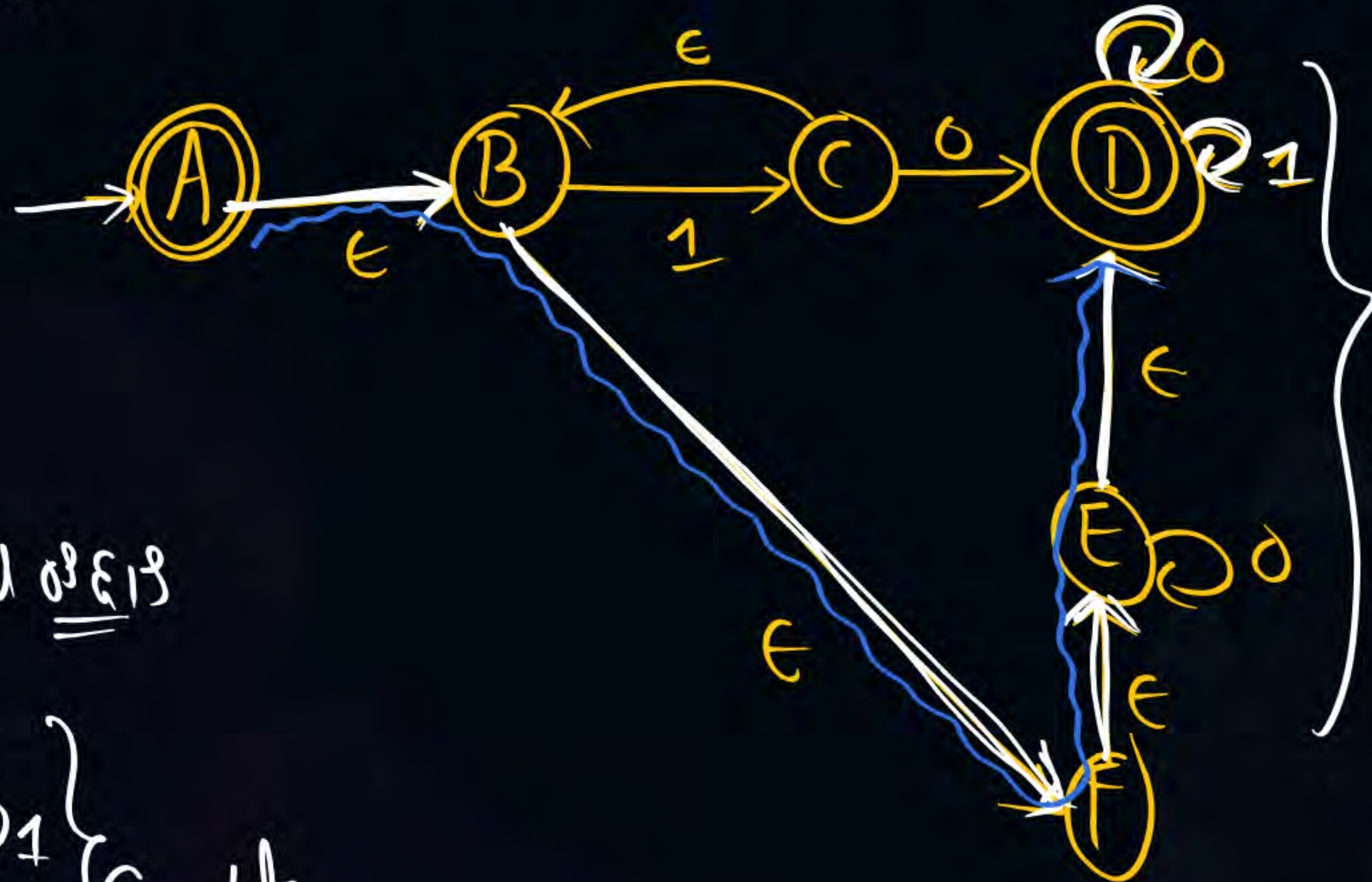
→ q ₀	{q ₀ , q ₁ , q ₂ }
q ₁	{q ₀ , q ₁ , q ₂ }
q ₂	{q ₀ , q ₁ , q ₂ }

min DFA



(Q) How many states are there in min DFA for given E-NFA

E-NFA

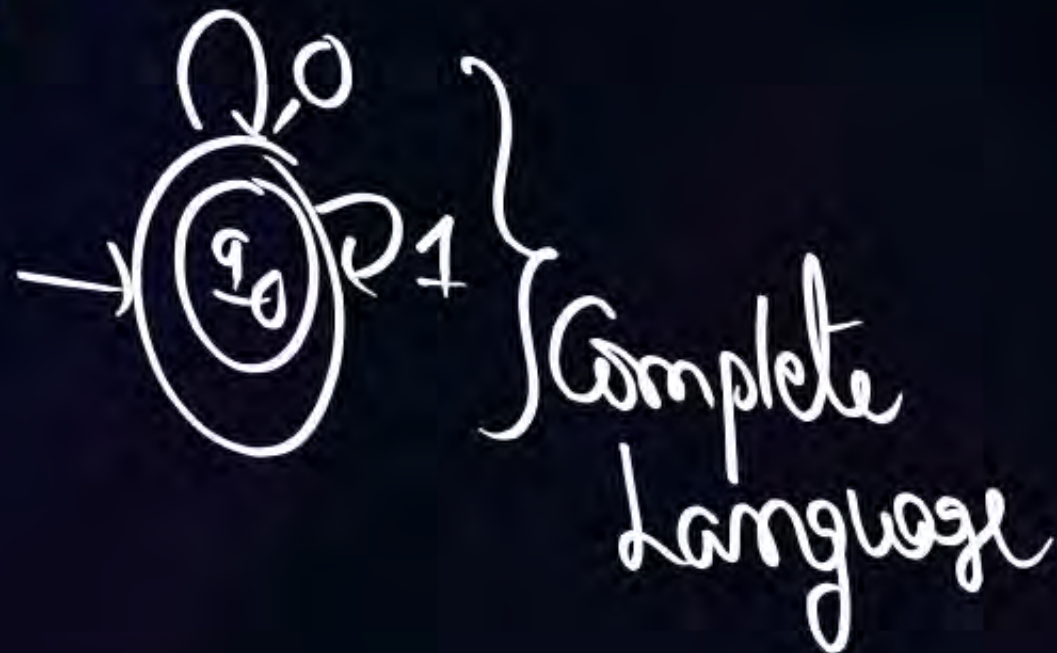


Complete Language

min DFA

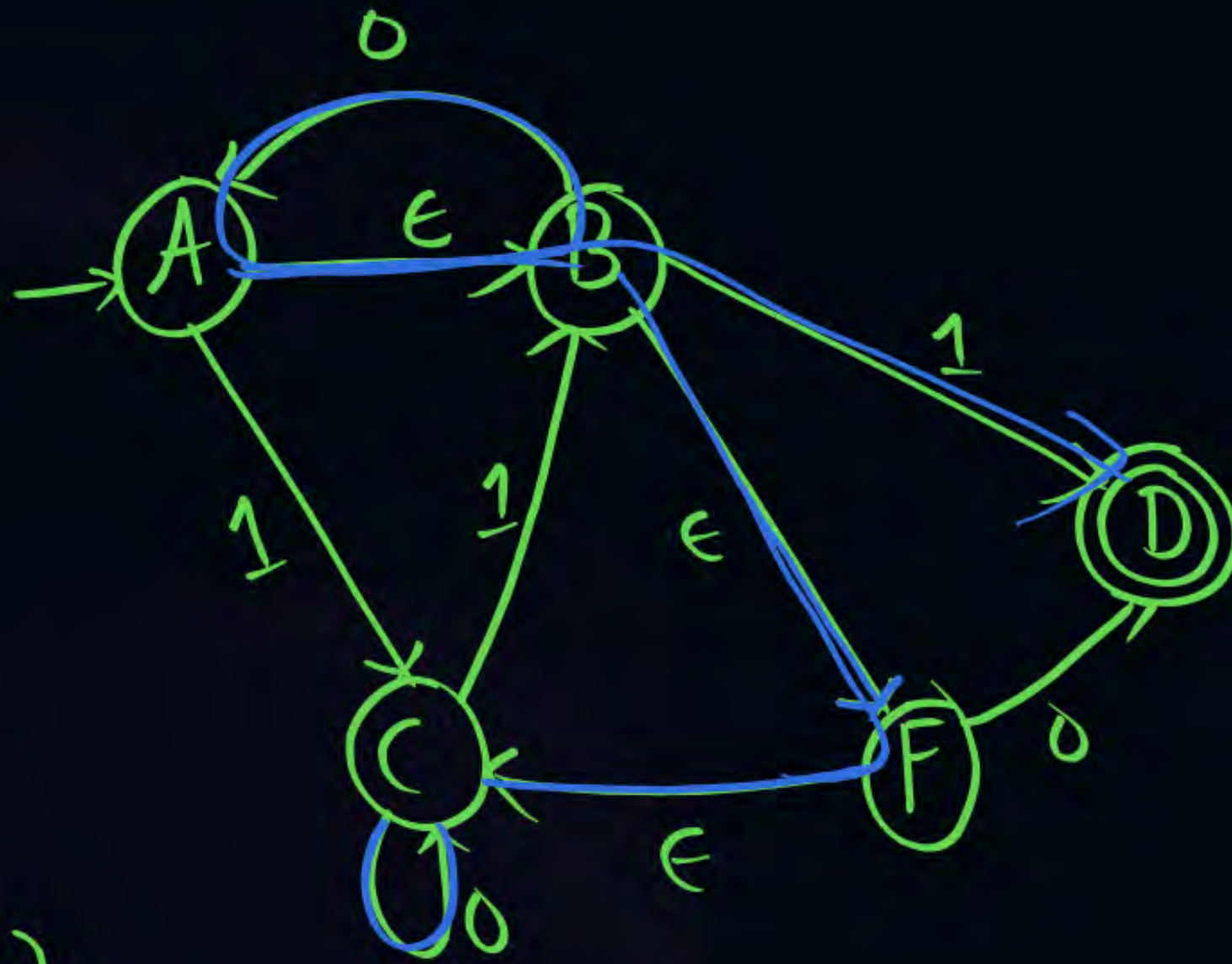
(1)

all 0's & 1's



(1)

(Q)



what is $\delta(A, 01) = ?$

$\{B, F, C, D\}$

(a) D

(b) $\{B, D\}$

(c) $\{B, C, D\}$

~~(d) $\{B, C, D, F\}$~~



Topic : Decision Properties of Finite Automata

{Algorithm exist}

Decidable problems

1. Equivalence Problem

2. Finiteness Problem

Loop (or) cycle exist $\rightarrow$ infinite.

otherwise finite

3. Emptiness Problem



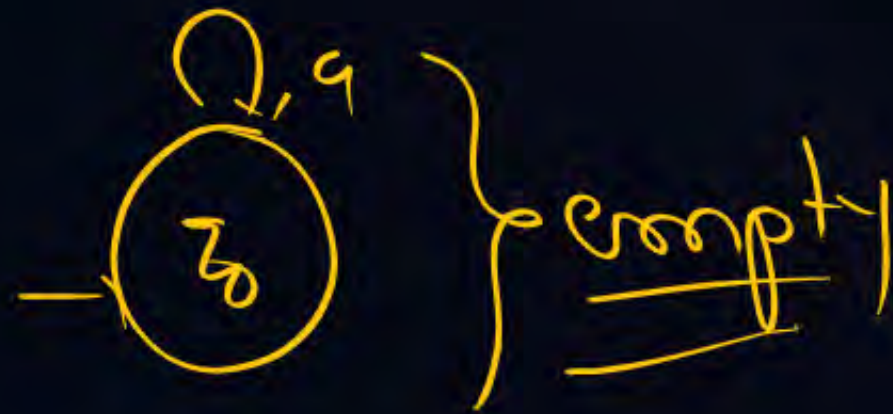
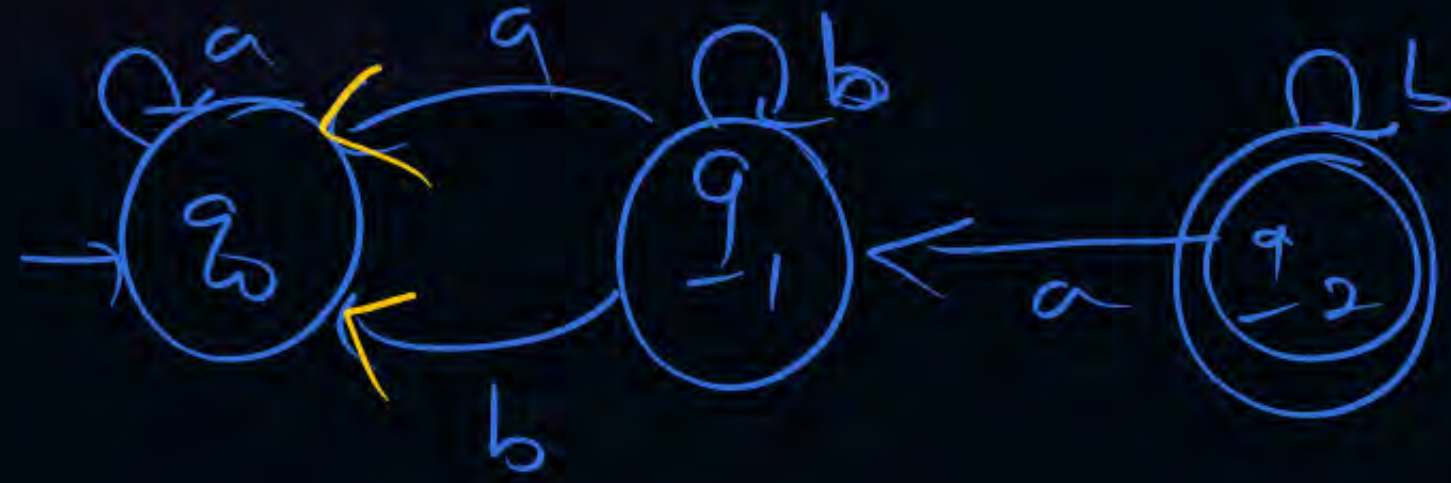
$\times \{ \epsilon \} \rightarrow$ finite language

- ① Eliminate inaccessible states
- ② Eliminate states from which final state is not reachable
- ③ In the resultant automata any loop (or) cycle exist then accepts infinite language



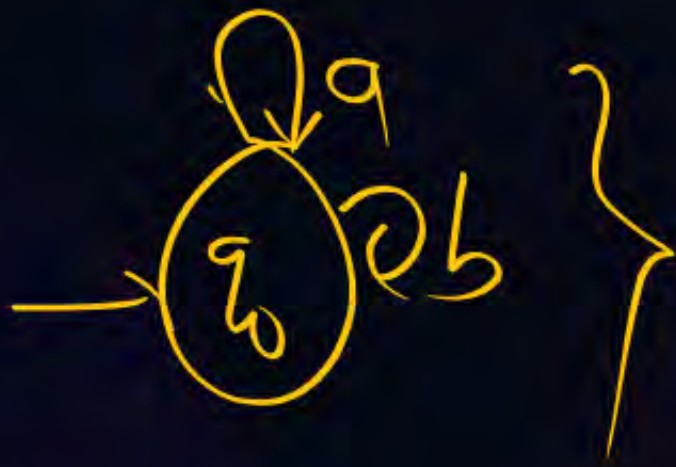
② $\{b, ab\}$

③ finite Language

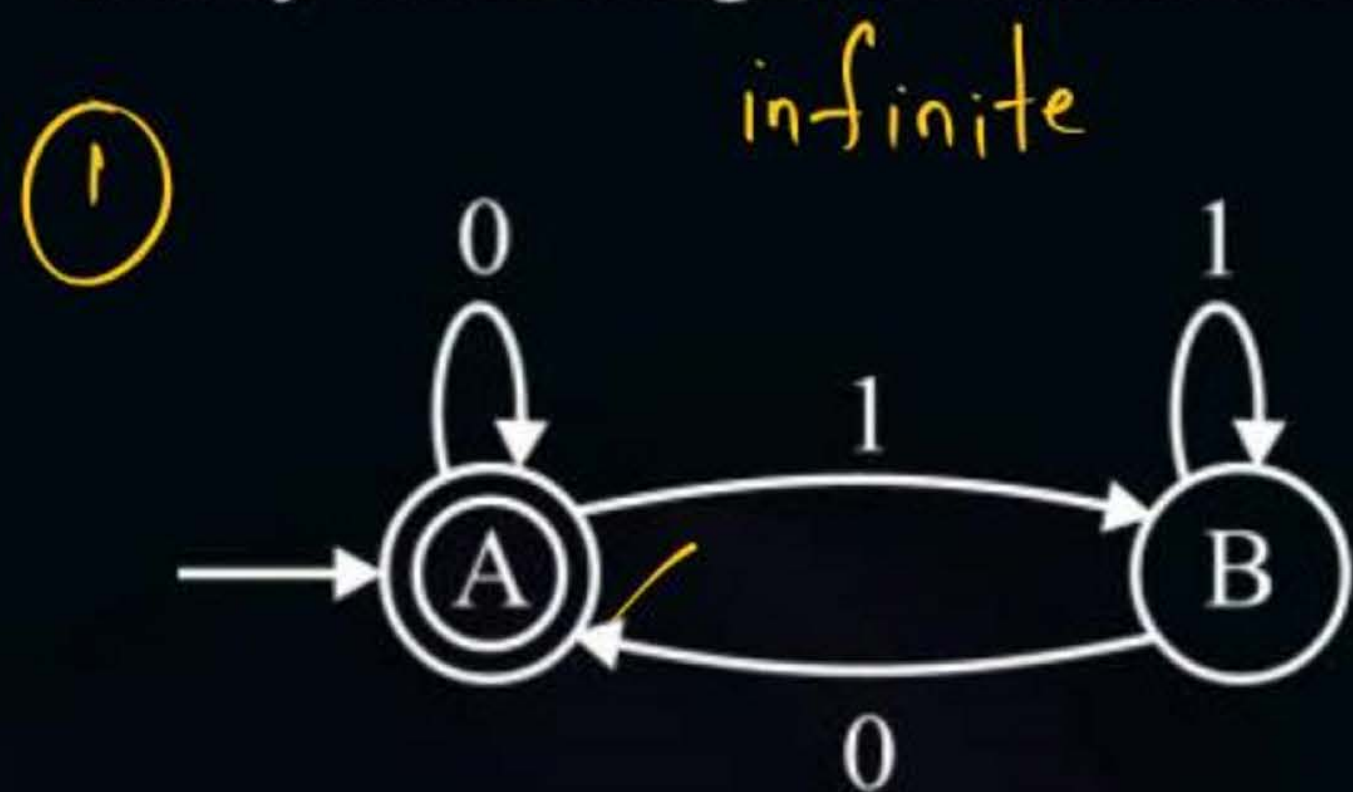


eliminate
① In accessible state

② final state $\rightarrow$ non empty
No final state $\rightarrow$ empty



#Q. Verify following two automata accepts same language or not

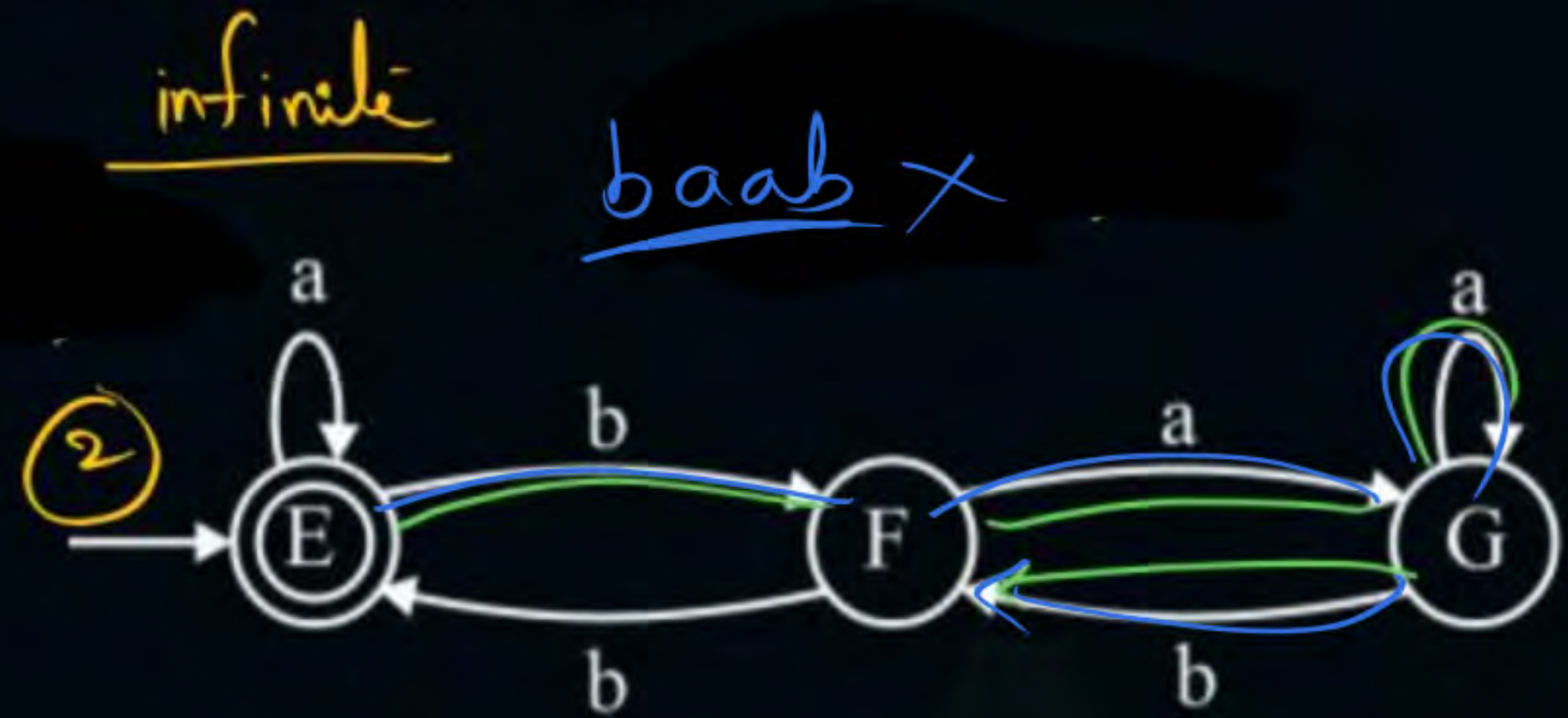
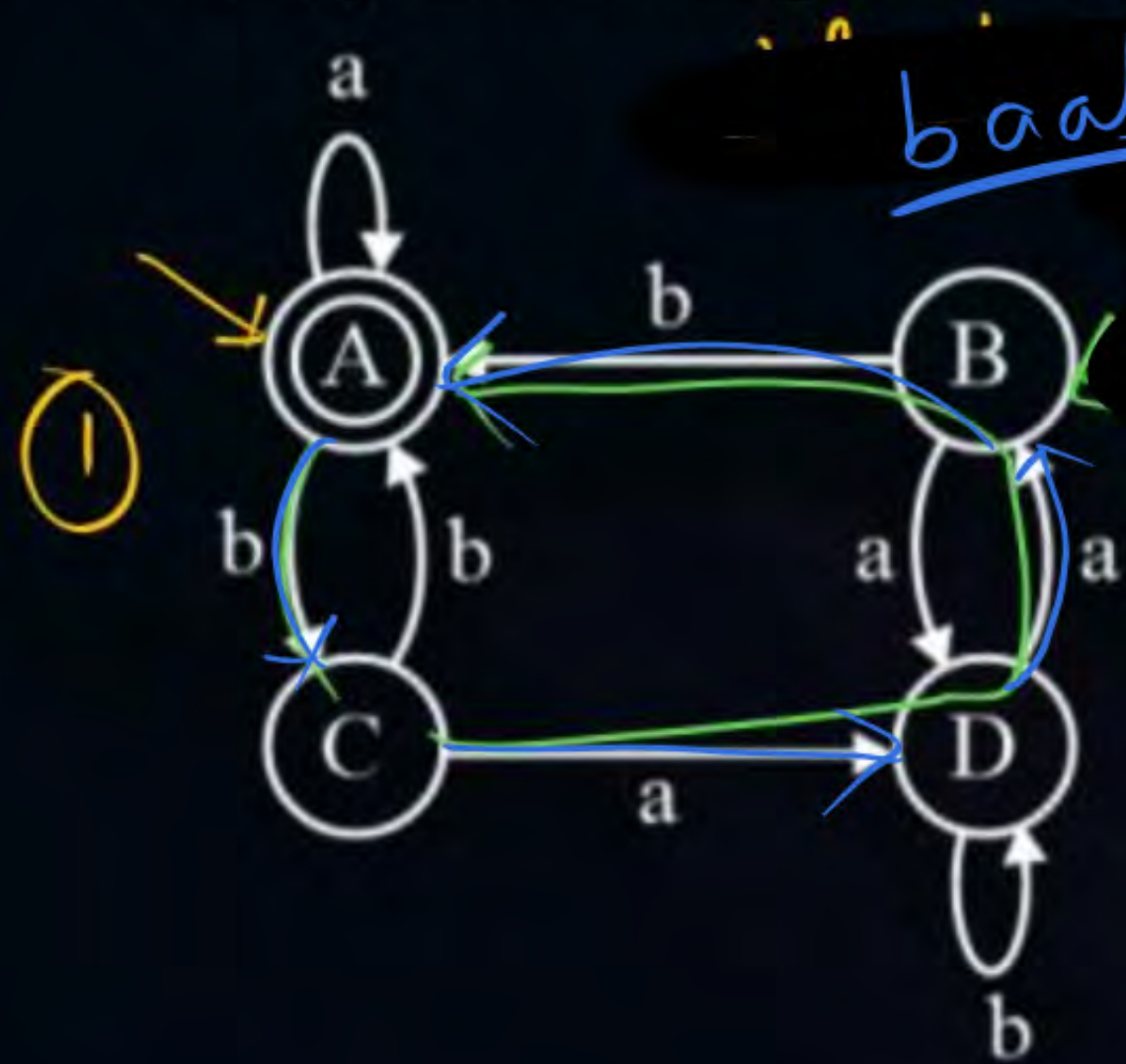


=
equal

Comparison table

	0	1
(A, C)	(A, D) ✓	(B, E) ✓
(A, D)	(A, D) ✓	(B, E) ✓
(B, E)	(A, C) ✓	(B, E) ✓

#Q. Verify following two automata accepts same language or not



	a	b
→ (A, E)	(A, E) ✓	(C, F) ✓
(C, F)	(D, G) ✓	(A, E) ✓
(D, G)	(B, G)	(D, F)
(B, G)	(D, G)	(A, F)

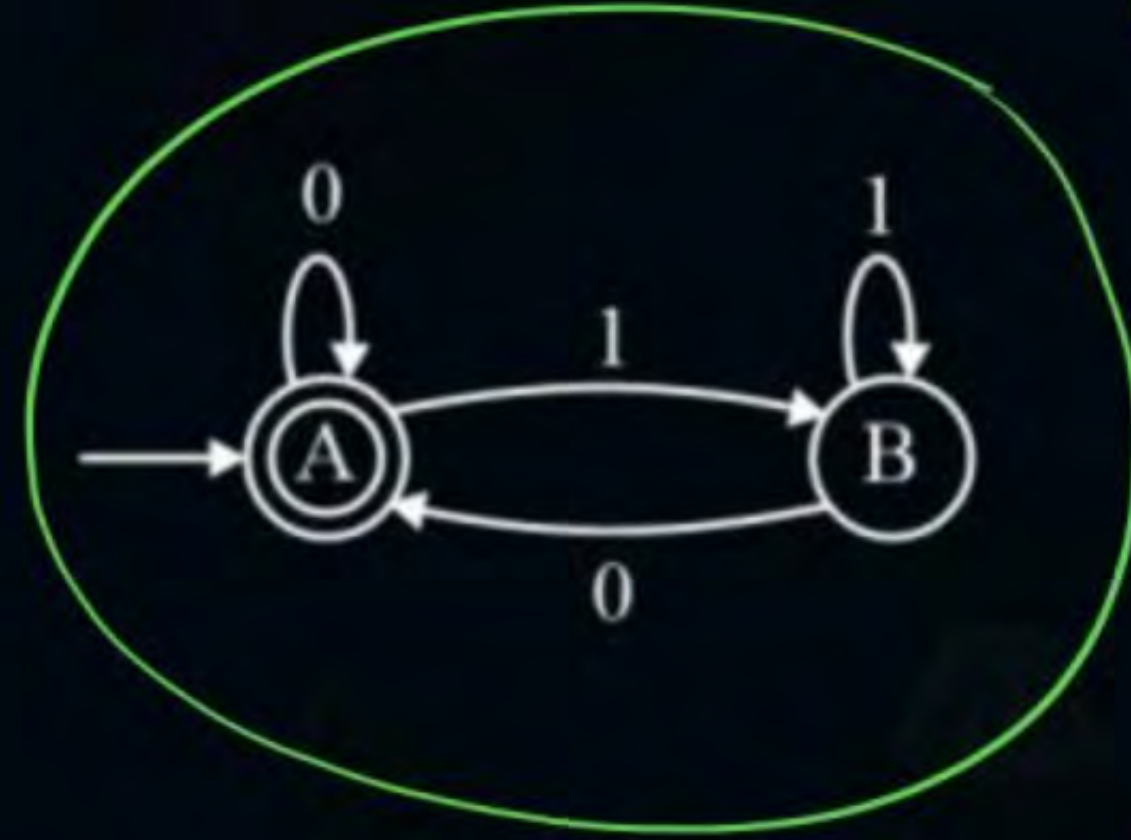
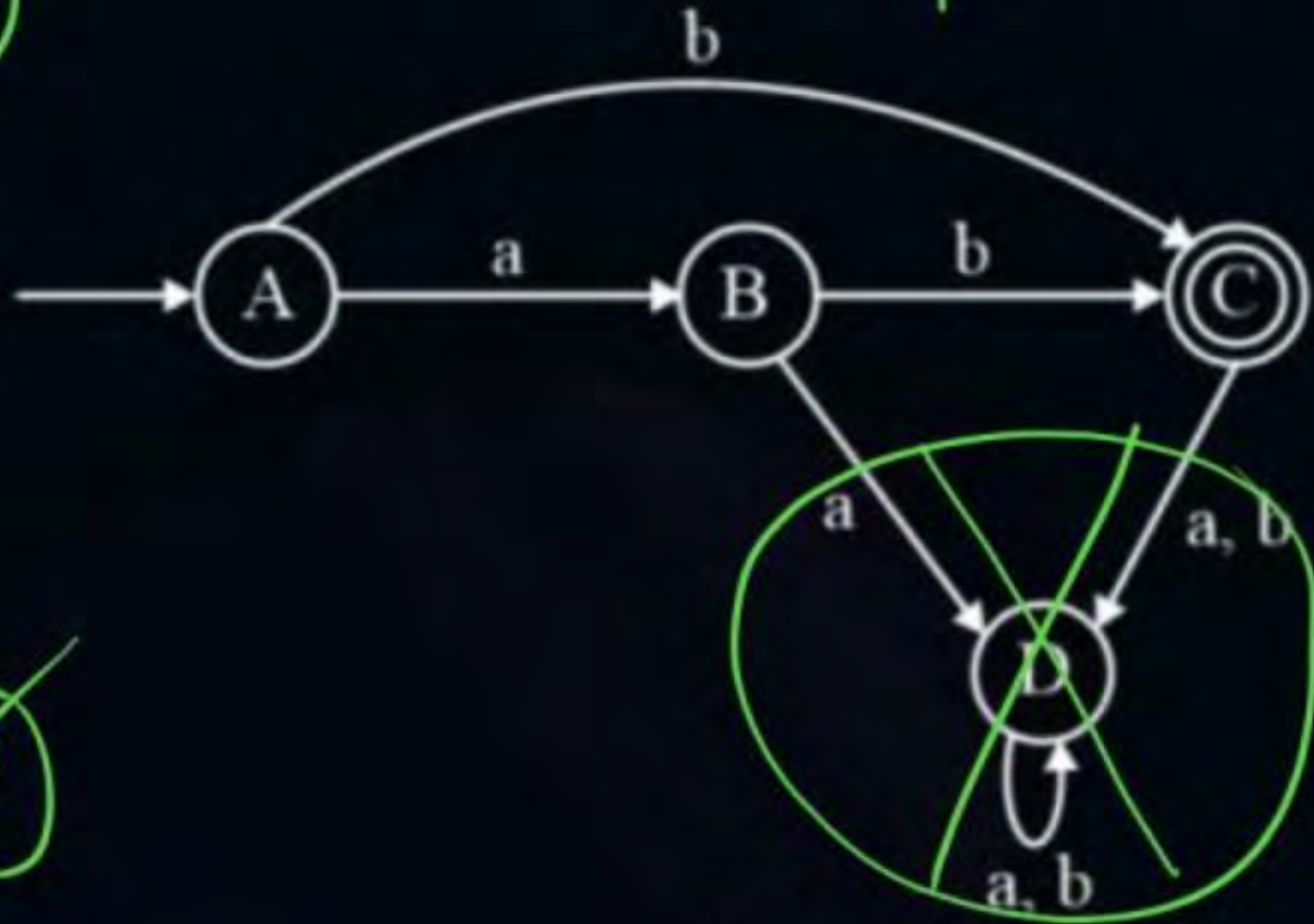
F NFA

Loop (or) Cycle exist or not?

#Q. Which of the following automata accepts infinite Language

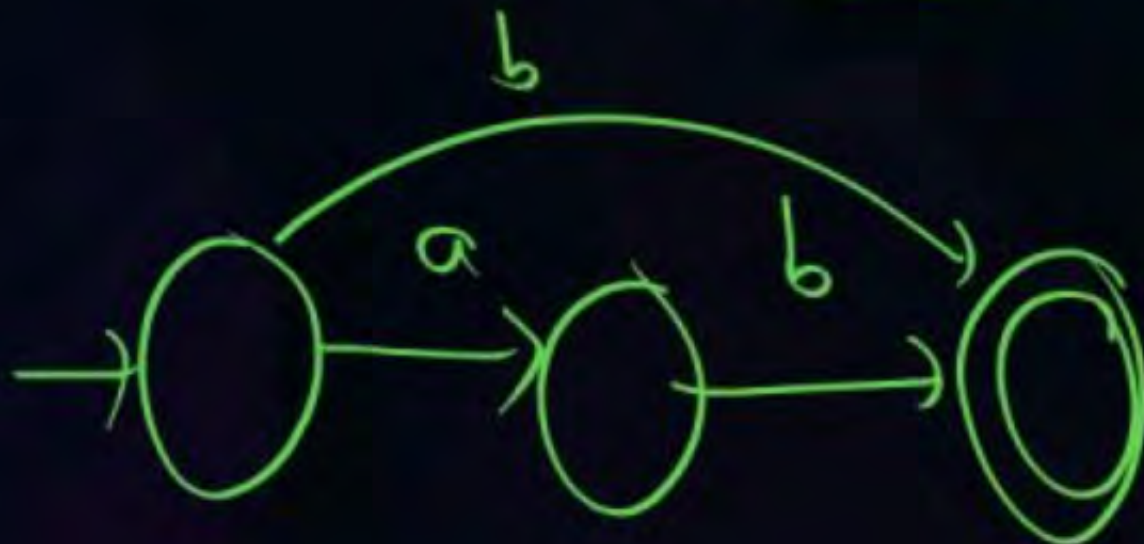
a) finite

b) infinite



infinite

$\{b, ab\}$



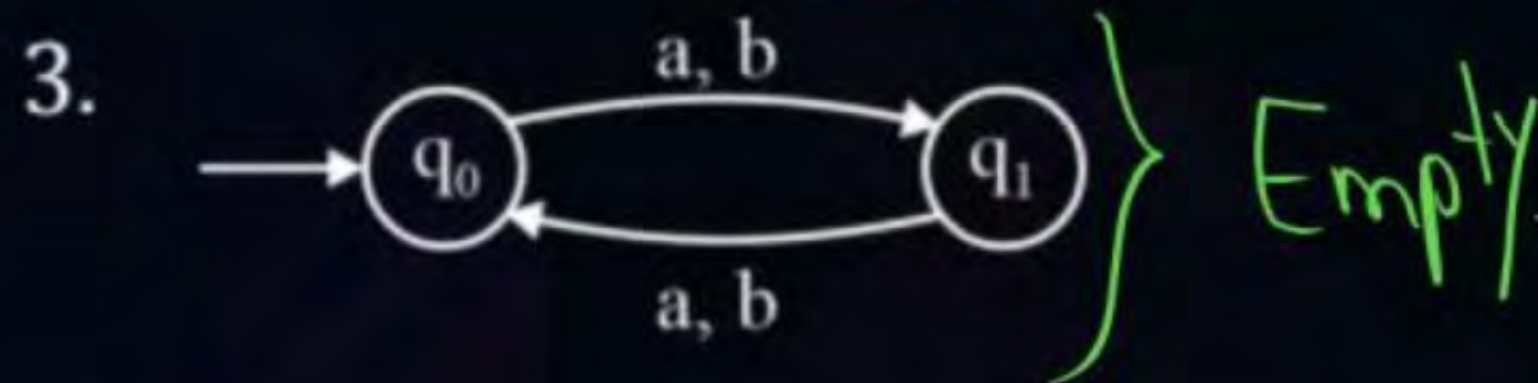
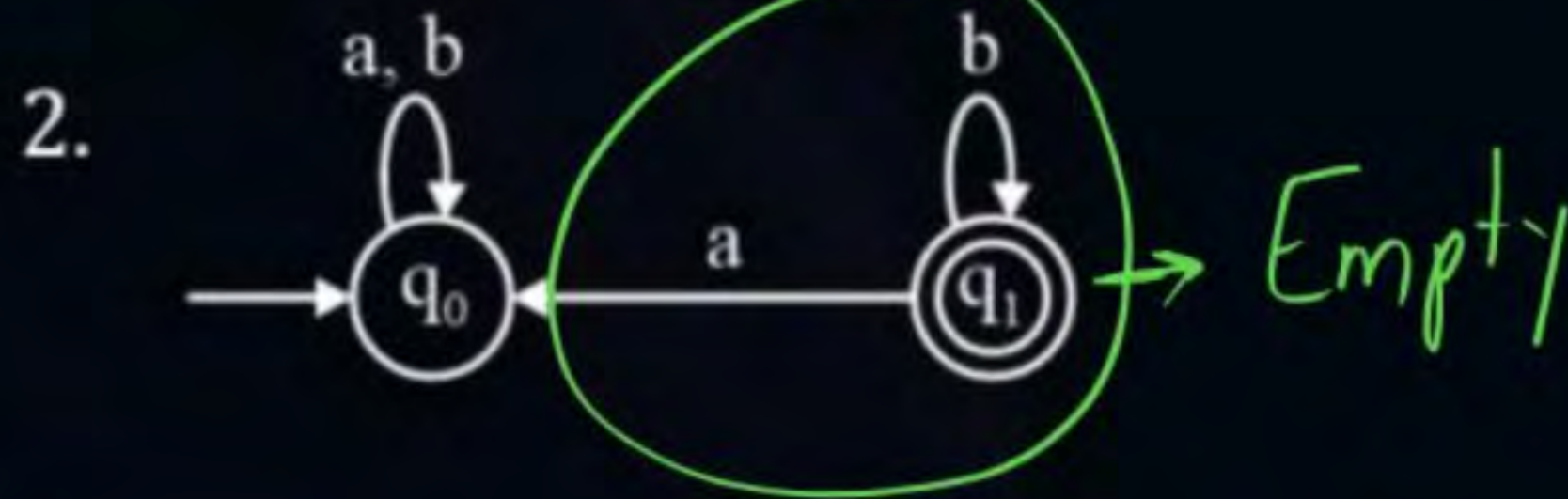
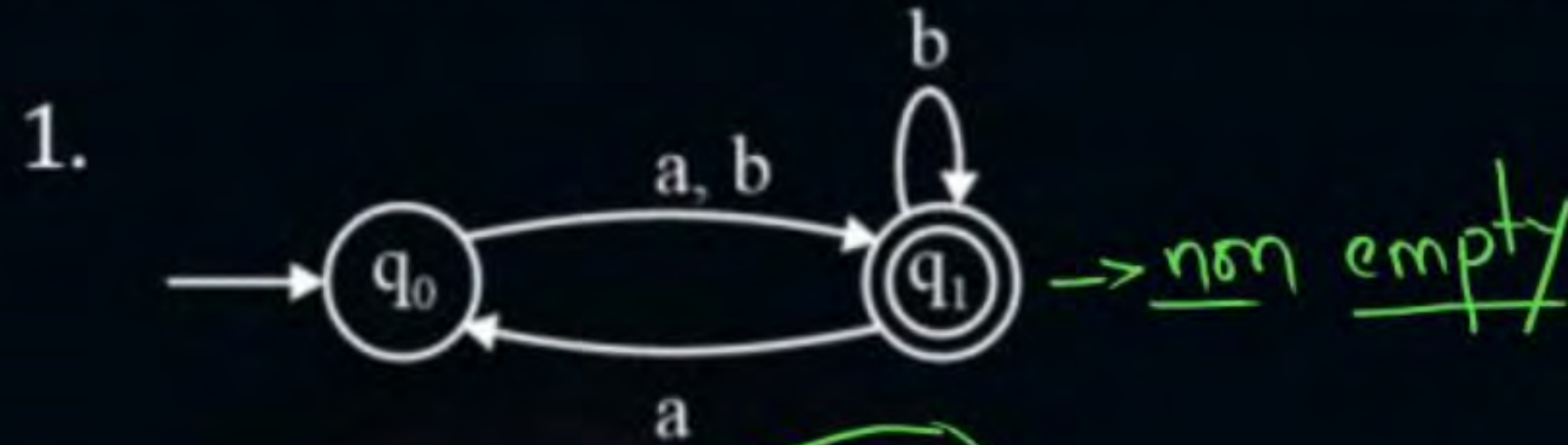
2

1

① Eliminate unreachable states

② Eliminate states from which final state is not reachable.

#Q. Which of the following automata accepts Empty Language.



① Eliminate unreachable states

② if no final state then empty language



THANK - YOU